

# Professional Software Engineering Portfolio Website: Design Specification

**Document purpose:** Software engineering design specification for your personal virtual CV and portfolio website.

**Deployment platform:** Netlify.

**Technology stack:** HTML5, CSS3, and JavaScript.

## 1. Project Overview

The Professional Software Engineering Portfolio Website is a single-page personal portfolio presenting your academic achievements as a final-year BSc Information Technology student at North West University, alongside your technical skills, leadership and employment experience, awards, selected projects, and direct contact options. It complements your curriculum vitae by providing an accessible, interactive overview for recruiters, lecturers, and prospective employers. The website is built as a static front-end application using HTML5, CSS3, and vanilla JavaScript, featuring a custom **Yellow and Black** theme, responsive design, semantic HTML, and a direct mailto: contact mechanism.

## 2. Project Purpose

The purpose of the website is to:

- Establish a professional online presence supporting internship, graduate, and software engineering applications.
- Showcase selected software engineering and game development projects (such as *The Harmoni Application VIP Project*, *Dragon Journey*, and *Lerato Orphanage System*), technical competencies, and academic development.
- Demonstrate practical front-end development capability through a responsive, maintainable website.
- Provide direct access to your downloadable CV, GitHub, LinkedIn, YouTube, Instagram, Twitch profiles, and professional email contact.
- Provide a central, easily updated location for professional information and future portfolio work.

## 3. Target Audience

The website is intended for:

- Software engineering and IT recruiters.
- Potential employers and internship coordinators.
- University lecturers and assessors.

- Professional networking contacts and North West University alumni.
- Fellow students, gamers, and project collaborators.

#### 4. Functional Requirements

The portfolio website provides the following functions:

- **Home:** Displays a concise professional introduction, career focus, typewriter text effect, social media icons, and a call-to-action to download your CV.
- **About:** Presents your professional biography, strategic mindset, Nondenominational Christian values, education history (Rabboni Christian School), languages, hobbies, and awards (Golden Key International Society, Cybersecurity certification, and Game Jam placement).
- **Projects:** Features detailed cards for *The Harmoni Application VIP Project (Android Studio)*, *Dragon Journey (48-hour Game Jam 2nd Place)*, and *Lerato Orphanage System (Oracle SQL Server)*.
- **Skills:** Presents your technical stack including Java (OOP, Data Structures, SQL), C# (Forms & ASP.NET), Python, HTML & CSS, and Microsoft Office tools using custom percentage skill bars.
- **Experience:** Details your extensive leadership and work history, including roles as Secretary General for the Democratic Alliance Student Organization, Secretary for the Geekulcha Student Society (NWU Chapter), News Reader for NWU RiverFM, Part-time Model for AGFS Models, Pageant Contestant, Legal Officer Faculty Student Leader, and Part-time Mechanic/Receptionist at Series Auto Clinic.
- **Contact:** Provides direct social links (LinkedIn, GitHub, YouTube, Instagram, Twitch) and a direct mailto: button that instantly opens the visitor's email client addressed to njcherane@proton.me.
- **Navigation:** Features a sticky desktop navigation bar and a JavaScript-controlled mobile hamburger menu supporting smooth anchor scrolling across all sections.

#### 5. Non-Functional Requirements

- **Performance:** Loads efficiently via lightweight static files and optimized local assets.
- **Responsiveness:** Adapts layout, typography, and containers seamlessly across desktop, laptop, tablet, and mobile breakpoints.
- **Browser Compatibility:** Functions cleanly across modern browsers (Google Chrome, Microsoft Edge, Mozilla Firefox, Safari).

- **Usability:** Maintains a clear visual hierarchy, high contrast, readable text, and immediate access to key links.
- **Accessibility & Maintainability:** Uses structured semantic elements, clean code separation (index.html, style.css, script.js), and consistent modular formatting.

## 6. Website Structure & Navigation

- **Single-Page Flow:** Home → About → Projects → Skills → Experience → Contact.
- **Interactive Navigation:** Desktop links stay fixed with active-state tracking; mobile layouts collapse into an interactive hamburger drawer via script.js.

## 7. Folder Structure

Plaintext

portfolio-website/

├— index.html

├— style.css

├— script.js

├— main.jpeg

└— Nyengedzo Jonathan Cherane's CV.pdf

## 8. Colour Palette & Typography

- **Primary Theme:** High-contrast **Yellow and Black** aesthetic (#FFCC00 accent color against #080808 dark background and #131313 secondary surfaces) optimized for a modern "Pro Coder" look.
- **Typography:** Utilizes the Google Font **Poppins** for clean, modern headings, subheadings, and UI elements, supported by system sans-serif fallbacks.

## 9. Technology Stack

- **Front-End:** HTML5, CSS3 (Flexbox, custom properties, responsive media queries), and Vanilla JavaScript (typewriter effect, mobile toggle, scroll spy).
- **Deployment Platform:** Netlify (for instant drag-and-drop or continuous git deployment and public portfolio hosting).